

Randomised clinical trial: gluten may cause depression in subjects with non-coeliac gluten sensitivity - an exploratory clinical study.

BACKGROUND: Current evidence suggests that many patients with self-reported non-coeliac gluten sensitivity (NCGS) retain gastrointestinal symptoms on a gluten-free diet (GFD) but continue to restrict gluten as they report 'feeling better'. **AIM:** To investigate the notion that a major effect of gluten in those with NCGS is on mental state and not necessarily on gastrointestinal symptoms. **METHODS:** Twenty-two subjects (24-62 years, five male) with irritable bowel syndrome who had coeliac disease excluded but were symptomatically controlled on a GFD, undertook a double-blind cross-over study. Participants randomly received one of three dietary challenges for 3 days, followed by a minimum 3-day washout before crossing over to the next diet. Challenge gluten-free food was supplemented with gluten (16 g/day), whey (16 g/day) or not supplemented (placebo). End-points included mental state as assessed by the Spielberger State Trait Personality Inventory (STPI), cortisol secretion and gastrointestinal symptoms. **RESULTS:** Gluten ingestion was associated with higher overall STPI state depression scores compared to placebo [M = 2.03, 95% CI (0.55-3.51), P = 0.010] but not whey [M = 1.48, 95% CI (-0.14 to 3.10), P = 0.07]. No differences were found for other STPI state indices or for any STPI trait measures. No difference in cortisol secretion was identified between challenges. Gastrointestinal symptoms were induced similarly across all dietary challenges. **CONCLUSIONS:** Short-term exposure to gluten specifically induced current feelings of depression with no effect on other indices or on emotional disposition. Gluten-specific induction of gastrointestinal symptoms was not identified. Such findings might explain why patients with non-coeliac gluten sensitivity feel better on a gluten-free diet despite the continuation of gastrointestinal symptoms. © 2014 John Wiley & Sons Ltd.